```
#FAKE NEWS DETECTION
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
import re
import string
```

```
df_fake=pd.read_csv("fake.csv")
df_true=pd.read_csv("true.csv")
```

```
df_fake.head()
```

	title	text	subject	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017

```
df_true.head()
```

	title	text	subject	date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017

```
df_fake.shape,df_true.shape
```

```
((23481, 4), (21417, 4))
```

```
df_fake["class"]=0
df_true["class"]=1
```

```
df_fake_manual_testing=df_fake.tail(10)
df_fake=df_fake.iloc[:-10]
df_true_manual_testing=df_true.tail(10)
df_true=df_true.iloc[:-10]
```

```
df_fake_manual_testing=pd.concat([df_fake_manual_testing,df_true_manual_testing],axis=0)
df_fake_manual_testing.to_csv("manual_testing.csv")
```

```
df_merge=pd.concat([df_fake,df_true],axis=0)
df_merge.head(10)
```

	title	text	subject	date	class
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017	0
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017	0
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017	0
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017	0
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017	0
5	Racist Alabama Cops Brutalize Black Boy While...	The number of cases of cops brutalizing and ki...	News	December 25, 2017	0
[illegible]	[illegible]	[illegible]	[illegible]	December 23, [illegible]	[illegible]

```
df=df_merge.drop(["title","subject","date"],axis=1)
df.head(10)
```

	text	class
0	Donald Trump just couldn t wish all Americans ...	0
1	House Intelligence Committee Chairman Devin Nu...	0
2	On Friday, it was revealed that former Milwauk...	0
3	On Christmas day, Donald Trump announced that ...	0
4	Pope Francis used his annual Christmas Day mes...	0
5	The number of cases of cops brutalizing and ki...	0
6	Donald Trump spent a good portion of his day a...	0
7	In the wake of yet another court decision that...	0
8	Many people have raised the alarm regarding th...	0
9	Just when you might have thought we d get a br...	0

```
df=df.sample(frac=1)
df.head(10)
```

	text	class
5931	Friday was Sasha Obama s 15th birthday. All ar...	0
1342	It was the cringe-inducing handshake seen an...	0
18887	DUBAI/PARIS (Reuters) - Saudi Arabia s decisio...	1
17221	MOGADISHU (Reuters) - Medics who rushed to hel...	1
5800	A racist Trump supporter who couldn t hack it ...	0
12173	WASHINGTON (Reuters) - North Korea s ambassado...	1
7301	NEW YORK/WASHINGTON (Reuters) - Hillary Clinto...	1
16499	VIENNA (Reuters) - The head of the United Nati...	1
535	WASHINGTON (Reuters) - U.S. Representative Joh...	1
13128	SYDNEY (Reuters) - Australia, concerned about ...	1

```
df.isnull().sum()
```

	0
text	0
class	0

dtype: int64

```
def word_drop(text):
  text=text.lower()
  text=re.sub(r'\[.*?\]','',text)
  text=re.sub(r"\W"," ",text)
  text=re.sub(r'https?://\S+|www\.\S+', '', text)
  text=re.sub(r'<.*?>+','',text)
  text=re.sub(r'[%s]' % re.escape(string.punctuation), '', text)
  text=re.sub(r'\n', '', text)
  text=re.sub(r'\w*\d\w*', '', text)
  return text
```

```
df["text"]=df["text"].apply(word_drop)
```

```
df.head(10)
```

	text	class
5931	friday was sasha obama s birthday all around...	0
1342	it was the cringe inducing handshake seen an...	0
18887	dubai paris reuters saudi arabia s decisio...	1
17221	mogadishu reuters medics who rushed to hel...	1
5800	a racist trump supporter who couldn t hack it ...	0
12173	washington reuters north korea s ambassado...	1
7301	new york washington reuters hillary clinto...	1
16499	vienna reuters the head of the united nati...	1
535	washington reuters u s representative joh...	1
13128	sydney reuters australia concerned about ...	1

```
x=df["text"]
y=df["class"]
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.25)
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
vectorization=TfidfVectorizer()
xv_train=vectorization.fit_transform(x_train)
xv_test=vectorization.transform(x_test)
```

```
#LOGISTIC REGRESSION
from sklearn.linear_model import LogisticRegression
LR=LogisticRegression()
LR.fit(xv_train,y_train)
```

```
▾ LogisticRegression
LogisticRegression()
```

```
LR.score(xv_test,y_test)
```

```
0.987338386090058
```

```
pred_LR=LR.predict(xv_test)
print(classification_report(y_test,pred_LR))
```

```
              precision    recall  f1-score   support

           0       0.99      0.99      0.99      5827
           1       0.98      0.99      0.99      5388

    accuracy                           0.99     11215
   macro avg       0.99      0.99      0.99     11215
weighted avg       0.99      0.99      0.99     11215
```

```
#Decision Tree Classification
from sklearn.tree import DecisionTreeClassifier
DT=DecisionTreeClassifier()
DT.fit(xv_train,y_train)
```

```
▾ DecisionTreeClassifier
DecisionTreeClassifier()
```

```
DT.score(xv_test,y_test)
```

```
0.9955416852429781
```

```
pred_DT=DT.predict(xv_test)
print(classification_report(y_test,pred_DT))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      5827
           1       1.00      0.99      1.00      5388

    accuracy                           1.00     11215
   macro avg       1.00      1.00      1.00     11215
weighted avg       1.00      1.00      1.00     11215
```

```
#RANDOM FOREST CLASSIFIER
from sklearn.ensemble import RandomForestClassifier
RFC=RandomForestClassifier(random_state=0)
RFC.fit(xv_train,y_train)
```

```
▾ RandomForestClassifier
RandomForestClassifier(random_state=0)
```

```
RFC.score(xv_test,y_test)
```

```
0.9913508693713776
```

```
pred_RFC=RFC.predict(xv_test)
print(classification_report(y_test,pred_RFC))
```

```
              precision    recall  f1-score   support

           0       0.99      0.99      0.99      5827
           1       0.99      0.99      0.99      5388

    accuracy                           0.99     11215
   macro avg       0.99      0.99      0.99     11215
weighted avg       0.99      0.99      0.99     11215
```

```
news="KILLED: Rancher and protest spokesman Robert  Lavoy  Finicum (Image Source: RT International)21st Century Wire says M
manual_testing(news)
```

```
LR prediction: Fake News
DT prediction: Fake News
RFC prediction: Fake News
```

```
#MANUAL TESTING
def output_lable(n):
  if n==0:
    return "Fake News"
  elif n==1:
    return "Not A Fake News"
def manual_testing(news):
  testing_news={"text":[news]}
  new_def_test=pd.DataFrame(testing_news)
  new_def_test["text"]=new_def_test["text"].apply(word_drop)
  new_x_test=new_def_test["text"]
  new_xv_test=vectorization.transform(new_x_test)
  pred_LR=LR.predict(new_xv_test)
  pred_DT=DT.predict(new_xv_test)
  pred_RFC=RFC.predict(new_xv_test)
  return print("\n\nLR prediction: {} \nDT prediction: {} \nRFC prediction: {}".format(output_lable(pred_LR[0]),output_lab
```

```
news="Patrick Henningsen 21st Century Wire UPDATE: 1-30-2016 Last night, the standoff at Malheur National Wildlife Refuge (
manual_testing(news)
```

```
LR prediction: Fake News
DT prediction: Fake News
RFC prediction: Fake News
```

```
news="LONDON (Reuters) - Abdul Daoud spilt most of the cappuccino into the saucer the first time he served Princess Diana,
manual_testing(news)
```

```
LR prediction: Not A Fake News
DT prediction: Not A Fake News
RFC prediction: Not A Fake News
```